Let $A \in \mathbb{R}^{m \times n}$ with rank r. $\exists$ orthogonal U & V and diagonal like $\Sigma \in \mathbb{R}^{m \times n}$ with

$$\Sigma = \left[\begin{array}{c|c} \Sigma_r & 0 \\ \hline 0 & 0 \end{array}\right] \quad \text{where} \quad \Sigma_r = \begin{bmatrix} \sigma_1 & & 0 \\ & \sigma_2 & \\ 0 & & \ddots \\ & & & \sigma_r \end{bmatrix} \text{ with}$$

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0 \quad \text{and}$$

$$A = U \Sigma V^T$$

Store $m \times n$ entries

$$= U_r \Sigma_r V_r^T$$

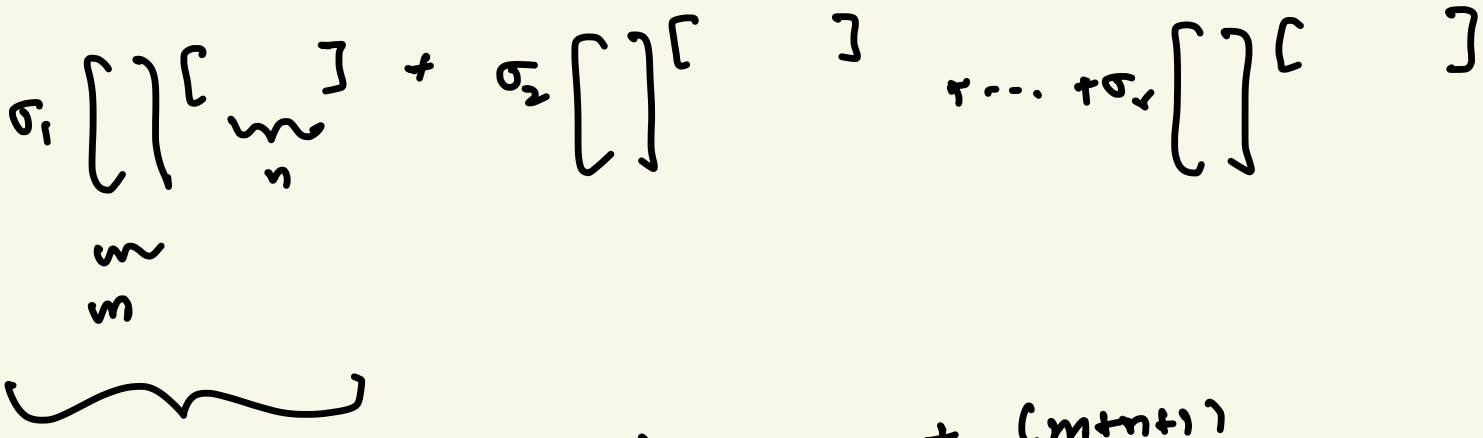
where $U_r = \begin{bmatrix} | & | & & | \\ u_1 & u_2 & \cdots & u_r \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{m \times r}$

$V_r = \begin{bmatrix} | & | & & | \\ v_1 & v_2 & \cdots & v_r \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times r}$

$$= \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T$$

$$= \sigma_1 \begin{bmatrix} | \\ u_1 \\ | \end{bmatrix} \begin{bmatrix} - & v_1^T & - \end{bmatrix} + \sigma_2 \begin{bmatrix} | \\ u_2 \\ | \end{bmatrix} \begin{bmatrix} - & v_2^T & - \end{bmatrix} + \cdots + \sigma_r \begin{bmatrix} | \\ u_r \\ | \end{bmatrix} \begin{bmatrix} - & v_r^T & - \end{bmatrix}$$

In order to store A, we need mn entries. (spaces)



$(m+n+1) + (m+n+1) + \cdots + (m+n+1)$

$= r(m+n+1)$

+ matrix multiplications & matrix additions

If r is very small,

$r(m+n+1) < mn$

10^4 is $11000 \approx 10(100+1000+1)$ 10^5

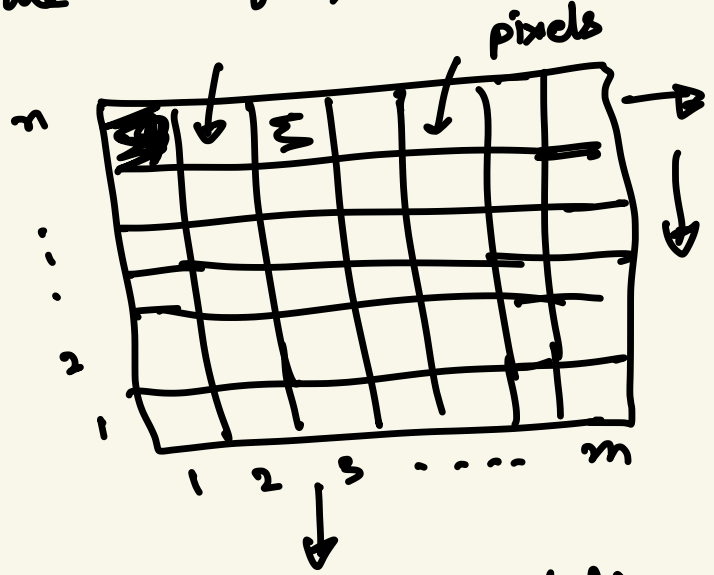
$m = 10^2 = 100$

$n = 10^3 = 1000$

$r = 10$

Applications :

Grey scale image / coloured image.



matrix representation

$$A = \begin{bmatrix} \bullet & \bullet & \bullet & \bullet & \bullet \\ & & & & \end{bmatrix}_{m \times n}$$

Grey scale image.

0 – 255

0, 1, 2, .., 255

white

black

$$A = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_r u_r v_r^T$$

$r = \text{rank}(A)$

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \cdots \geq \sigma_r > 0$$

(Eckart-Young) 1940's

$$B = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T + \cdots + \sigma_k u_k v_k^T$$

← low rank approximation of A

(In fact, B is the closest rank k matrix to A)

$\text{rank}(B) = k < r = \text{rank}(A)$

$k(m+n+1) < mn$

If k is very small.

(LORA) - 2022 onwards